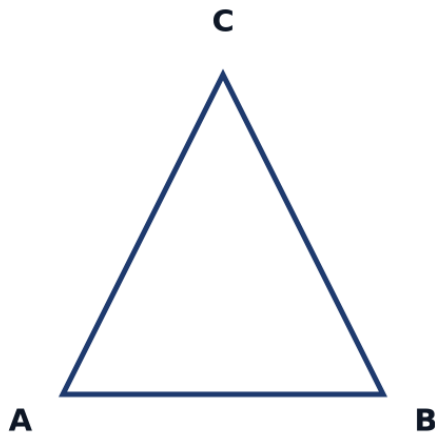


Advanced Coordinate Geometry



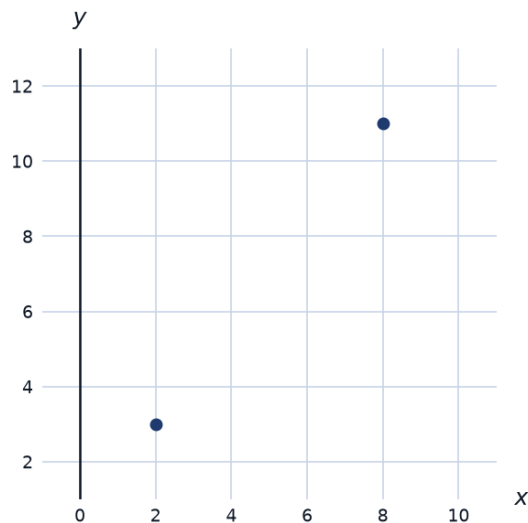
Distance, midpoint, and slope combined

- 1 Triangle ABC has vertices $A(-2, 1)$, $B(4, 1)$, and $C(1, 7)$. Find the length of the median from C to the midpoint of AB .



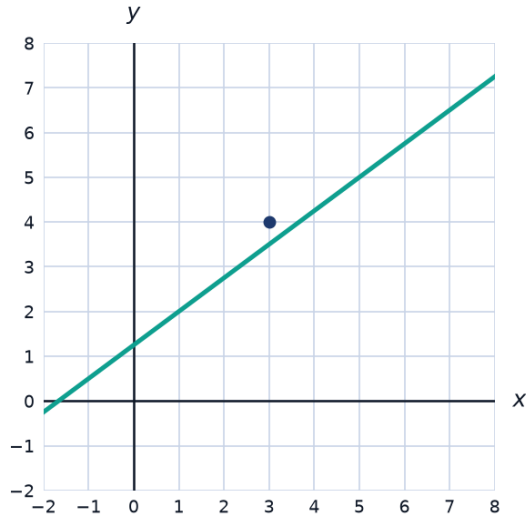
Equations of circles from given conditions

- 2 Find the equation of a circle with a diameter having endpoints $(2, 3)$ and $(8, 11)$.



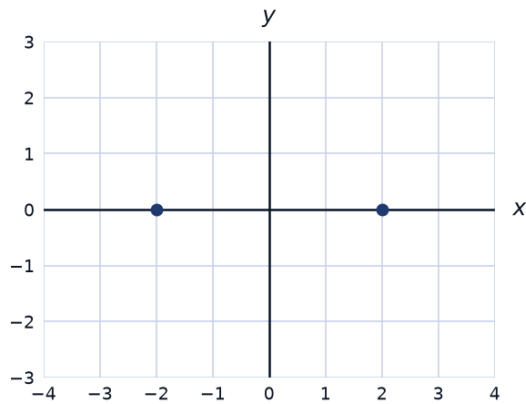
Distance from a point to a line

- 3 Find the distance from the point $(3, 4)$ to the line $3x - 4y + 5 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.



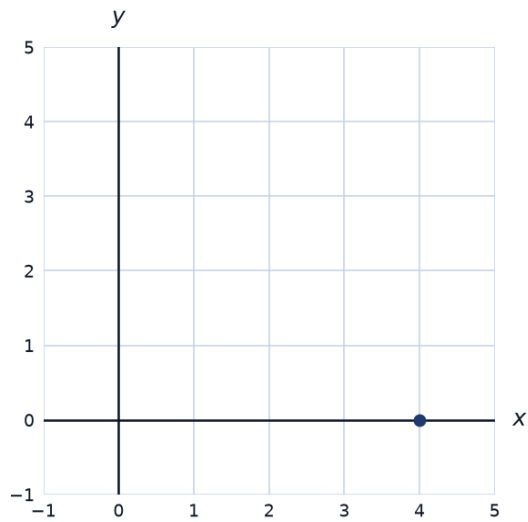
Locus problems

- 4 Find the equation of the locus of points equidistant from $(2, 0)$ and $(-2, 0)$.



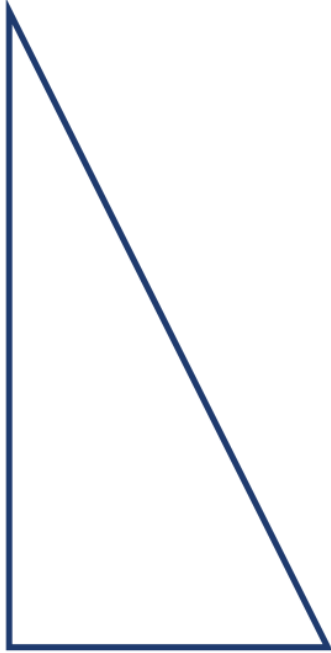
Rotations and general transformations

- 5 A point $(4, 0)$ is rotated 90° counterclockwise about the origin. Find the coordinates of the image point, using the rotation rule $(x, y) \rightarrow (-y, x)$.



Dilations and scale factor

- 6 A triangle with vertices $(1, 2)$, $(3, 2)$, $(1, 6)$ is dilated by a scale factor of 3 centered at the origin. Find the coordinates of the image vertices.



Finding the equation of a parabola from three points

- 7 A parabola $y = ax^2 + bx + c$ passes through $(0, 3)$, $(1, 6)$, and $(-1, 2)$. Find a , b , and c .

A multi-part triangle-in-the-plane problem

- 8 This question has two parts. Triangle POR has $P(0, 0)$, $O(6, 0)$, and $R(3, 6)$. (a) Find the area of the triangle using the coordinate formula $\text{Area} = \frac{1}{2}|x_P(y_Q - y_R) + x_Q(y_R - y_P) + x_R(y_P - y_Q)|$. (b) Verify the area using the base-height formula instead.

Finding the equation of a line given a point and a perpendicular condition

- 9 Find the equation of the perpendicular bisector of the segment connecting $(2, 3)$ and $(8, 7)$.
-

Translating and reflecting equations of curves

- 10 The graph of $y = x^2$ is reflected over the x -axis, then shifted 3 units up. Write the equation of the resulting graph.
-

Finding vertices of a polygon given transformations

- 11 A square has vertices $(0, 0)$, $(4, 0)$, $(4, 4)$, $(0, 4)$. Find the coordinates after translating the square 3 units right and 2 units down.
-

Finding the intersection of a line and a parabola

- 12 Find the points of intersection of $y = x^2 - 2x - 3$ and $y = x - 3$.
-

A multi-part quadrilateral classification problem

- 13 This question has three parts. Quadrilateral $WXYZ$ has vertices $W(0, 0)$, $X(6, 0)$, $Y(8, 4)$, $Z(2, 4)$. (a) Find the length of each side. (b) Find the slopes of WX and ZY , and of WZ and XY . (c) Based on parts (a) and (b), classify the quadrilateral as precisely as possible.
-

Finding the equation of a tangent line to a circle

- 14 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$, using the fact that the tangent is perpendicular to the radius at that point.
-

Finding the vertex and axis of symmetry of a parabola from three points

- 15 A parabola passes through $(-1, 0)$, $(3, 0)$, and $(1, -8)$. Find the axis of symmetry and the vertex.
-

Multiple choice — easy

16 Find the slope of the line through $(4, -5)$ and $(8, -2)$.

a. $-\frac{3}{4}$

b. $\frac{3}{4}$

c. $-\frac{4}{3}$

d. $\frac{4}{3}$

17 Find the slope of the line through $(3, -6)$ and $(2, 2)$.

a. $\frac{1}{8}$

b. 8

c. $-\frac{1}{8}$

d. -8

18 Find the slope of the line through $(-6, -5)$ and $(-7, 6)$.

a. -11

b. $-\frac{1}{11}$

c. 11

d. $\frac{1}{11}$

19 Find the slope of the line through $(-3, 8)$ and $(-9, 5)$.

a. 2

b. -2

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

20 Find the slope of the line through $(-4, 6)$ and $(5, 9)$.

a. -3

b. $\frac{1}{3}$

c. $-\frac{1}{3}$

d. 3

21 Find the slope of the line through $(-6, 6)$ and $(-5, 6)$.

a. 0

b. 1

c. 2

d. 3

22 Find the slope of the line through $(-9, -5)$ and $(-6, -8)$.

a. 1

b. 2

c. -1

d. 0

23 Find the slope of the line through $(-1, -9)$ and $(4, 0)$.

a. $-\frac{9}{5}$

b. $\frac{5}{9}$

c. $\frac{9}{5}$

d. $-\frac{5}{9}$

24 Find the slope of the line through $(9, 0)$ and $(-6, 4)$.

a. $-\frac{4}{15}$

b. $\frac{15}{4}$

c. $-\frac{15}{4}$

d. $\frac{4}{15}$

25 Find the slope of the line through $(-5, -1)$ and $(-4, 8)$.

a. -9

b. 9

c. $\frac{1}{9}$

d. $-\frac{1}{9}$

26 Find the slope of the line through $(9, -1)$ and $(-2, 9)$.

a. $\frac{10}{11}$

b. $-\frac{10}{11}$

c. $-\frac{11}{10}$

d. $\frac{11}{10}$

27 Find the slope of the line through $(-3, 7)$ and $(7, -4)$.

a. $-\frac{11}{10}$

b. $\frac{10}{11}$

c. $\frac{11}{10}$

d. $-\frac{10}{11}$

28 Find the slope of the line through $(-8, -1)$ and $(-1, 5)$.

a. $\frac{7}{6}$

b. $-\frac{6}{7}$

c. $-\frac{7}{6}$

d. $\frac{6}{7}$

29 Find the slope of the line through $(8, -9)$ and $(6, 4)$.

a. $\frac{2}{13}$

b. $\frac{13}{2}$

c. $-\frac{2}{13}$

d. $-\frac{13}{2}$

30 Find the slope of the line through $(-9, -3)$ and $(-7, 1)$.

a. 2

b. -2

c. $-\frac{1}{2}$

d. $\frac{1}{2}$

Multiple choice — medium

31 Find the distance from the point $(9, -1)$ to the line $20x - 21y - 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.

a. $\frac{248}{29}$

b. $\frac{219}{29}$

c. -190

d. $\frac{190}{29}$

- 32 Find the distance from the point $(1, -1)$ to the line $3x - 4y + 6 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{18}{5}$
 - b. $\frac{13}{5}$
 - c. $\frac{23}{5}$
 - d. -13
- 33 Find the distance from the point $(8, -6)$ to the line $5x + 12y - 4 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $-\frac{36}{17}$
 - b. $\frac{36}{17}$
 - c. $\frac{36}{13}$
 - d. $-\frac{36}{13}$
- 34 Find the distance from the point $(0, 3)$ to the line $8x + 15y - 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{70}{17}$
 - b. $\frac{36}{17}$
 - c. $\frac{53}{17}$
 - d. $\frac{36}{23}$
- 35 Find the distance from the point $(9, 7)$ to the line $5x + 12y + 11 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{166}{13}$
 - b. $\frac{140}{17}$
 - c. $\frac{140}{13}$
 - d. $\frac{153}{13}$

- 36 Find the distance from the point $(3, -8)$ to the line $12x + 35y - 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- $-\frac{253}{37}$
 - $\frac{253}{37}$
 - $-\frac{253}{47}$
 - $\frac{253}{47}$
- 37 Find the distance from the point $(-2, -2)$ to the line $7x - 24y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- $-\frac{19}{17}$
 - $\frac{44}{25}$
 - $\frac{69}{25}$
 - $\frac{19}{25}$
- 38 Find the distance from the point $(8, 0)$ to the line $12x - 35y - 8 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- $\frac{162}{37}$
 - $\frac{125}{37}$
 - $\frac{88}{37}$
 - $-\frac{88}{23}$
- 39 Find the distance from the point $(-3, 5)$ to the line $12x + 35y - 15 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- $\frac{161}{37}$
 - $\frac{124}{47}$
 - $\frac{198}{37}$
 - $\frac{124}{37}$

- 40 Find the distance from the point $(-8, 0)$ to the line $9x + 40y + 6 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $-\frac{66}{49}$
 - b. $\frac{66}{49}$
 - c. $-\frac{66}{41}$
 - d. $\frac{66}{41}$
- 41 Find the distance from the point $(-7, -8)$ to the line $12x - 35y + 10 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{280}{37}$
 - b. $\frac{243}{37}$
 - c. $\frac{206}{37}$
 - d. $-\frac{206}{23}$
- 42 Find the distance from the point $(-5, 5)$ to the line $8x + 15y - 13 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{39}{17}$
 - b. $\frac{22}{23}$
 - c. $\frac{22}{17}$
 - d. $\frac{56}{17}$
- 43 Find the distance from the point $(7, 1)$ to the line $3x + 4y + 9 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{44}{5}$
 - b. $\frac{39}{5}$
 - c. $\frac{34}{7}$
 - d. $\frac{34}{5}$

- 44 Find the distance from the point $(2, -3)$ to the line $20x + 21y + 7 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{16}{41}$
 - b. $-\frac{16}{41}$
 - c. $-\frac{16}{29}$
 - d. $\frac{16}{29}$
- 45 Find the distance from the point $(2, -2)$ to the line $5x - 12y + 12 = 0$, using $d = \frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}}$.
- a. $\frac{46}{13}$
 - b. $\frac{59}{13}$
 - c. $\frac{72}{13}$
 - d. $-\frac{46}{7}$
-

Multiple choice — hard

- 46 Find the y-intercept of the perpendicular bisector of the segment connecting $(-2, -1)$ and $(4, -6)$.
- a. $-\frac{47}{10}$
 - b. $-\frac{8}{3}$
 - c. $-\frac{37}{10}$
 - d. $\frac{47}{10}$
- 47 Find the y-intercept of the perpendicular bisector of the segment connecting $(-5, -6)$ and $(0, 0)$.
- a. $\frac{61}{12}$
 - b. 0
 - c. $-\frac{61}{12}$
 - d. $-\frac{49}{12}$

48 Find the y-intercept of the perpendicular bisector of the segment connecting $(8, 7)$ and $(6, -5)$.

a. $-\frac{13}{6}$

b. -41

c. $\frac{13}{6}$

d. $\frac{19}{6}$

49 Find the y-intercept of the perpendicular bisector of the segment connecting $(2, -3)$ and $(8, 7)$.

a. -5

b. 5

c. 6

d. $-\frac{19}{3}$

50 Find the y-intercept of the perpendicular bisector of the segment connecting $(-6, 0)$ and $(6, -3)$.

a. $-\frac{3}{2}$

b. $\frac{1}{2}$

c. $\frac{3}{2}$

d. $-\frac{1}{2}$

51 Find the sum of the x-coordinates of the intersection points of $y = x^2 - 12$ and $y = -4x$.

a. -12

b. -4

c. -2

d. -6

- 52 Find the sum of the x -coordinates of the intersection points of $y = x^2 - 4$ and $y = -4x + 1$.
- a. -2
 - b. -5
 - c. -6
 - d. -4
- 53 Find the sum of the x -coordinates of the intersection points of $y = x^2 + 2x + 21$ and $y = -6x + 9$.
- a. -10
 - b. -6
 - c. -8
 - d. 12
- 54 Find the sum of the x -coordinates of the intersection points of $y = x^2 + 4x - 11$ and $y = -x - 5$.
- a. -3
 - b. -5
 - c. -6
 - d. -7
- 55 Find the sum of the x -coordinates of the intersection points of $y = x^2 + 15x + 28$ and $y = 4x - 2$.
- a. -9
 - b. -13
 - c. 30
 - d. -11

- 56 A parabola $y = ax^2 + bx + c$ passes through $(0, 5)$, $(1, 8)$, and $(-1, -4)$. Find a .
- a. -3
 - b. 6
 - c. -2
 - d. -4
- 57 A parabola $y = ax^2 + bx + c$ passes through $(0, 5)$, $(1, 2)$, and $(-1, 2)$. Find a .
- a. -4
 - b. -2
 - c. 0
 - d. -3
- 58 A parabola $y = ax^2 + bx + c$ passes through $(0, -3)$, $(1, -1)$, and $(-1, -11)$. Find a .
- a. -4
 - b. -3
 - c. -2
 - d. 5
- 59 A parabola $y = ax^2 + bx + c$ passes through $(0, 7)$, $(1, 0)$, and $(-1, 12)$. Find a .
- a. -6
 - b. -1
 - c. 0
 - d. -2

60 A parabola $y = ax^2 + bx + c$ passes through $(0, -3)$, $(1, -2)$, and $(-1, -2)$. Find a .

a. 1

b. 2

c. 0

d. 3